

Houston Dataset:

1. Description

This dataset includes a hyperspectral image along with a LiDAR-derived digital surface model (DSM), both collected by the ITRES CASI-1500 sensor in June 2012. The scene covers the University of Houston campus and its surrounding urban area in Texas, USA, with a ground sampling distance (GSD) of 2.5 m. The image size is 349×1905 pixels, and the hyperspectral data contain 144 bands spanning the 380–1050 nm wavelength range with a spectral resolution of 10 nm. For land-use and land-cover (LULC) classification, the scene is annotated at the pixel level into 15 distinct classes. In this work, we follow the official training and testing split provided by the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest, which ensures a well-separated partition as illustrated in the figure below. The class names and the number of samples for each class are summarized in Table 1.

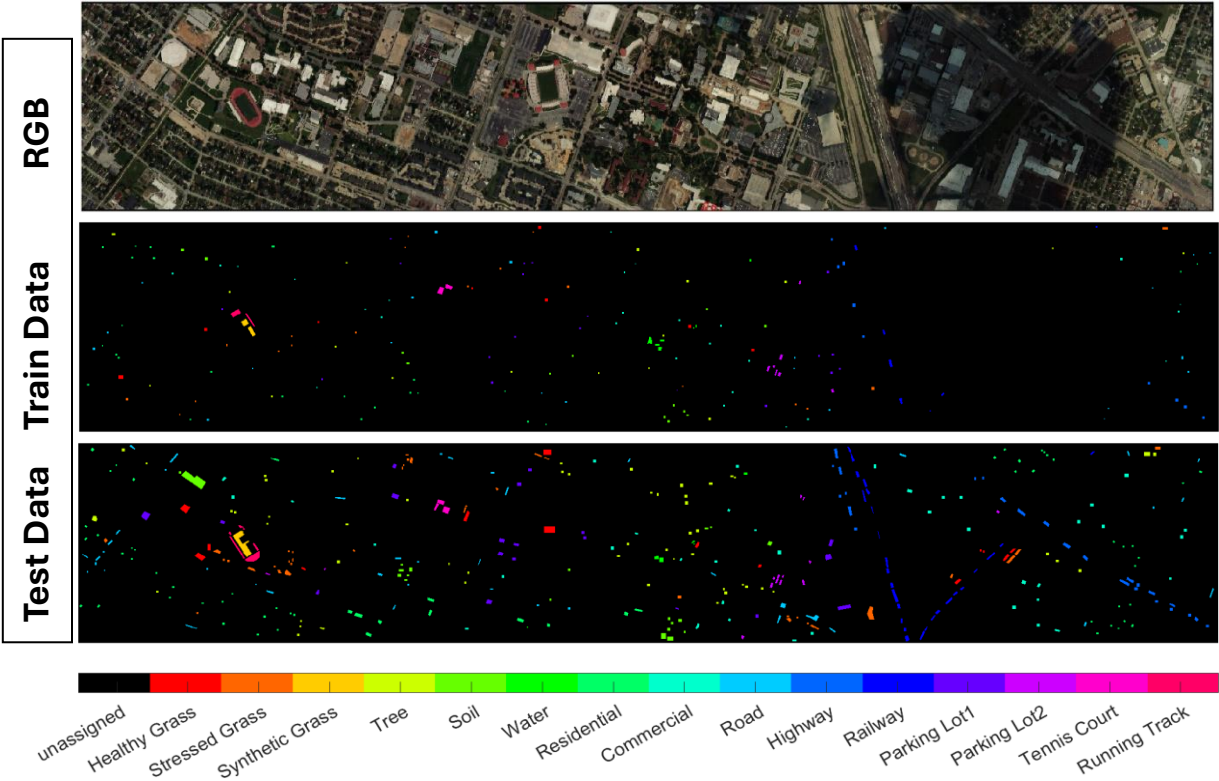


Table 1: Description of the Houston2013 HS–LiDAR Dataset with Ground Object Types and Associated Training/Testing Samples

Class No.	Class Name	Training Set	Testing Set
1	Healthy Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot1	192	1041
13	Parking Lot2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
Total	–	2832	12197
Percentage	$2832/(2832 + 12497) = 18.84\%$		

2. Results

Upon applying the methods used in the paper, here are the results of each model qualitatively and quantitatively as shown in Table 2.

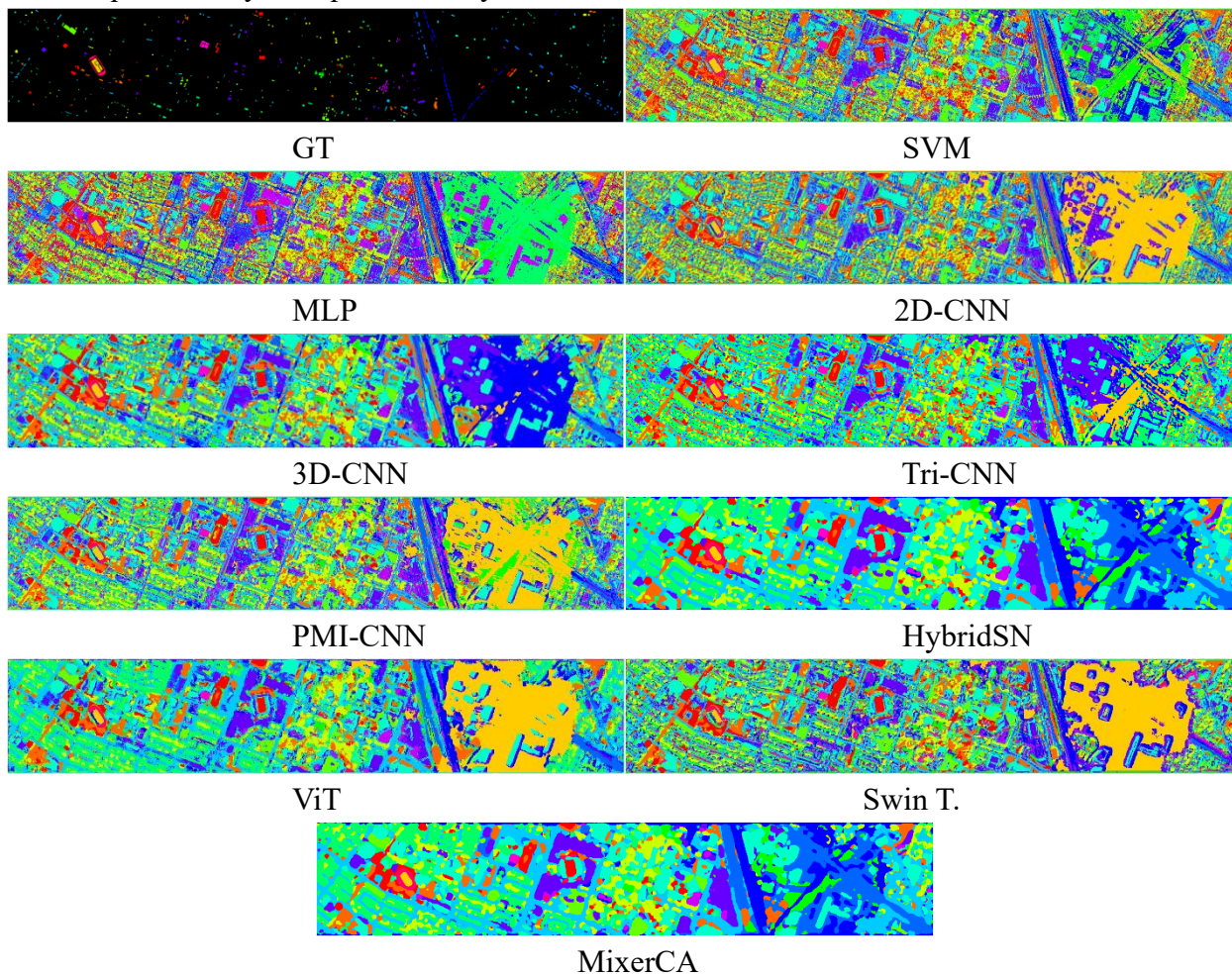


Table 2: classification results of Houston13 using the algorithms used in the paper. This is following the train/test split used in the referenced paper above.

Class	SVM	MLP	2D-CNN	3D-CNN	Tri-CNN	PMI-CNN	HybridS N	ViT	Swin T.	MixerC A
1	80.63	81.96	79.11	81.96	81.86	80.44	82.53	82.81	80.91	79.96
2	83.27	76.32	69.64	82.24	75.94	84.30	85.15	79.32	85.15	85.17
3	97.60	97.23	57.82	96.24	91.68	70.69	89.70	49.70	84.75	97.82
4	88.35	91.19	84.19	93.37	92.14	93.09	91.86	72.25	82.86	82.01
5	98.86	96.97	97.92	99.91	99.72	97.44	100.00	92.42	99.81	99.81
6	98.60	90.91	67.83	88.11	85.31	82.52	79.02	65.73	80.42	98.69
7	80.41	82.56	71.46	79.76	88.34	81.90	75.19	60.07	84.70	76.68
8	65.24	36.75	66.76	77.68	73.88	60.02	72.46	46.15	68.38	81.67
9	78.28	50.61	68.93	80.26	72.71	77.15	72.71	67.14	71.39	85.84
10	54.44	41.89	39.48	50.39	50.39	49.23	52.90	33.49	57.82	66.41
11	80.55	70.21	69.64	93.26	82.83	64.23	81.40	62.90	69.07	95.92
12	70.61	50.05	45.82	80.69	83.96	63.40	81.08	47.55	93.18	98.85
13	63.51	57.54	80.70	63.51	80.00	85.61	92.98	58.60	81.40	80.00
14	99.19	97.98	92.71	100.00	99.19	97.98	100.00	94.74	95.95	99.60
15	98.31	98.73	65.75	100.00	89.22	74.00	97.25	48.20	70.82	99.56
OA (%)	80.13	70.99	69.48	83.30	81.50	75.76	81.40	63.75	79.65	86.67
AA (%)	82.66	74.73	70.52	84.49	83.14	77.47	83.62	64.07	80.44	88.65
K x 100	78.51	68.62	67.00	81.86	79.93	73.85	79.82	60.88	77.99	85.54

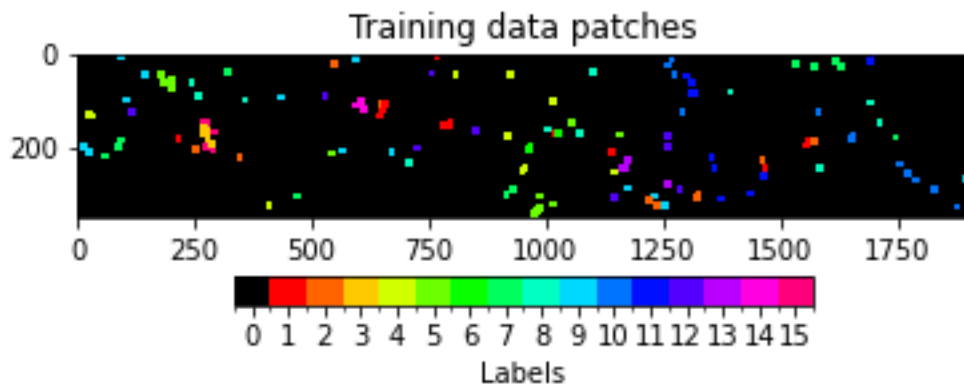
3. Discussion:

The comparative analysis across different models highlights clear differences in classification quality. Traditional approaches such as **SVM** and **MLP** exhibit noisy predictions and poor boundary preservation, particularly in urban classes like *Residential* and *Commercial*, where spectral similarity leads to confusion. **2D-CNN** improves spatial consistency but still suffers from mixed-pixel misclassifications, while **3D-CNN** demonstrates stronger spectral-spatial feature learning and produces smoother results with clearer object delineation. **Tri-CNN** and **PMI-CNN** enhance specific classes, yet both models introduce artifacts and inconsistencies in fine-structured regions. **HybridSN** performs competitively with balanced accuracy across most classes but struggles in areas with limited training data. Transformer-based models (**ViT** and **Swin Transformer**) capture global dependencies but underperform in heterogeneous urban scenes, producing fragmented maps and misclassification of smaller objects. Notably, all methods experience difficulties in the **cloud-shadow region** of the Houston dataset (visible in the RGB image), where spectral distortion reduces classification accuracy and increases class confusion. Despite these challenges, **MixerCA** achieved the highest overall accuracy, average accuracy, and Kappa score among the compared methods. Its classification map was less influenced by the cloudy region and it maintained good recognition of small object classes such as Tennis Court and Running Track. In addition, MixerCA obtained the best per-class accuracy in 8 out of the 15 categories, showing consistent improvements over other models.

In the other part of the experiments, the model will be trained with only 1% of data to maintain the same setup followed in the paper, Table 3 shows the train/test split for each class:

Table 3: Houston2013 HS–LiDAR Dataset with Ground Object Types and Associated Training/Testing Samples with only 1% of data is used for training

Class No.	Class Name	Training Set	Testing Set
1	Healthy Grass	13	1238
2	Stressed Grass	13	1241
3	Synthetic Grass	7	690
4	Tree	12	1232
5	Soil	12	1230
6	Water	3	322
7	Residential	13	1255
8	Commercial	12	1232
9	Road	13	1239
10	Highway	12	1215
11	Railway	12	1223
12	Parking Lot1	12	1221
13	Parking Lot2	5	464
14	Tennis Court	4	424
15	Running Track	7	653
Total	–	150	14879



And the above figure shows the training patches distribution in the image.

Table 4: classification results of Houston13 using the algorithms used in the paper. The train/test split was 1/99

Class	SVM	MLP	2D-CNN	3D-CNN	Tri-CNN	PMI-CNN	HybridSN	ViT	Swin T.	MixerCA
1	93.46	78.19	80.94	96.20	97.66	88.53	94.10	75.77	83.52	84.01
2	97.82	63.26	91.78	98.47	84.05	99.60	98.55	70.51	87.99	90.99
3	100.00	96.52	78.26	91.74	12.03	85.51	89.13	52.32	53.33	96.99
4	94.97	87.18	58.20	94.48	78.57	90.99	76.14	63.47	72.56	79.18
5	98.05	95.37	98.37	100.00	97.56	99.84	99.27	84.23	99.51	99.44
6	81.06	13.04	34.78	86.34	47.21	36.96	81.99	23.60	18.01	84.62
7	85.66	64.30	62.31	60.48	65.66	82.47	79.68	77.77	79.44	88.41
8	67.21	67.45	62.99	68.75	64.04	70.94	53.57	30.36	53.98	57.40
9	83.05	64.97	45.28	39.14	47.05	47.78	43.18	46.57	76.35	83.87
10	75.06	27.24	49.63	98.52	74.24	89.22	93.25	53.74	26.01	86.80
11	69.17	62.47	20.69	77.84	49.71	71.22	84.55	46.61	74.33	100.00
12	74.04	63.23	89.52	83.95	91.89	94.84	69.37	67.65	74.20	86.46
13	19.18	34.48	4.09	87.28	74.57	29.96	43.10	19.40	29.53	64.61
14	61.08	85.14	46.46	86.08	100.00	80.42	93.87	84.67	96.93	100.00
15	98.47	91.58	93.42	100.00	98.47	100.00	100.00	76.57	92.65	100.00
OA (%)	82.55	68.13	64.61	83.38	73.23	81.57	79.88	60.44	70.99	85.53
AA (%)	79.89	66.29	61.11	84.62	72.18	77.88	79.98	58.22	67.89	86.85
K x 100	81.11	65.53	61.67	82.04	71.15	80.05	78.25	57.20	68.58	84.44

When the training set was reduced to only 1% of the labeled samples, most deep learning models experienced a considerable performance drop, as expected under limited supervision. This decline was particularly pronounced for transformer-based models (ViT and Swin Transformer), which generally require larger datasets to generalize well. In contrast, the traditional SVM classifier maintained performance comparable to the previous setup with ~18% training data, highlighting its stability under small-sample conditions. Tri-CNN, owing to its large number of parameters, showed a more notable decrease in accuracy when trained on such a small subset. Despite the challenging setting, **MixerCA achieved the highest overall accuracy (85.53%), average accuracy (86.85%), and Kappa (84.44)**, outperforming all other methods and demonstrating greater robustness to limited training data.